Healthcare AI Strategy Consulting Case Study

Should a Mid Sized Hospital Invest in an AI Powered Clinical Deterioration Prediction System

Consolidated investment assessment and conditional pilot proposal
12 November 2026

Executive summary

Recommendation: do not approve hospital wide deployment on the current financial case. Consider a capped, controlled 12 week pilot on two internal medicine wards only after clinical, technical, governance and funding prerequisites are met. Use comparison wards and prespecified evaluation methods to reduce uncertainty before a further investment decision.

The hypothetical base case assumes 300 beds, 12,000 admissions and 240 unplanned ICU transfers per year. A 20% reduction, valued at €1,500 of avoidable variable cost per transfer, generates €72,000 in annual gross savings against €90,000 in annual recurring cost. The annual operating balance is −€18,000. Including €120,000 initial investment, the three year net benefit is −€174,000 and ROI is −44.6%.

The required three year break even reduction is 36.1%, above the 35% optimistic scenario. Patient safety and workflow benefits could still justify further evaluation, but neither their existence nor their monetary value has been demonstrated in this case. A pilot is an investment in learning, not proof of positive ROI.

Decision conditions

Authorize live use only with validated data feeds, a frozen versioned model, accountable clinical response, safe documentation, auditability, access control and a tested fallback. Approve a separate pilot budget and stop rules. A favorable pilot may justify a larger evaluation; it does not automatically authorize scale up.

Status of the analysis

This is a hypothetical educational consulting case. Financial inputs are assumptions, technology ratings are user judgments, and thresholds are illustrative proposals requiring local approval. No patient outcomes, vendor performance, realized savings or completed pilot results are claimed.

Report guide: problem and issue tree • IBM Lab 1 opportunity assessment • IBM Lab 2 transformation evaluation • financial model • decision and pilot • KPIs and safety gates • stakeholder requirements • MoSCoW priorities • appendices.

1 Problem and issue tree

The investment question

Hospital leadership must decide whether an AI system that predicts clinical deterioration can improve timely assessment and care sufficiently to justify its implementation burden, operating cost and clinical risk. The relevant intervention includes data collection, prediction, alert delivery, human response and documentation; prediction alone cannot produce patient benefit.

The proposed scope is adult internal medicine ward care, subject to a locally approved intended use and eligibility definition. The case does not specify a vendor, prediction horizon, clinical endpoint or deployed model. These must be defined before validation or live use. Existing escalation pathways remain active throughout the pilot.

Issue tree for the decision

Decision branch	Questions to resolve	Evidence required
Clinical effectiveness	Does the model identify relevant deterioration, and do alerts lead to timely, appropriate care?	Local validation, calibration, sensitivity, positive predictive value, response timestamps and reviewed outcomes.
Operational feasibility	Can staff respond reliably without excessive burden? Can data and alerts be delivered safely?	Workflow testing, alert burden, staffing coverage, training, data latency and downtime drills.
Financial viability	Do realizable incremental benefits exceed total implementation and operating costs?	Avoidable cost estimates, binding quotations, pilot resource use and scenario analysis.
Governance and acceptability	Can accountable and acceptable use be maintained?	Clinical safety review, privacy and security assessment, applicable approvals, patient communication and model change control.

The branches separate patient benefit, delivery feasibility, financial valuation and governance. Dependencies remain explicit: clinical actions create workload; workflow affects outcomes; finance values only validated, incremental benefits. Count the same benefit once.

Testable hypotheses

Clinical: alerts shorten time to first doctor contact and appropriate action without delaying necessary escalation. Operational: the response pathway remains reliable within agreed staff capacity. Financial: a credible range of sustained effect and realizable cost avoidance supports an acceptable investment case. Each hypothesis is unresolved.

Outcome interpretation

Unplanned ICU transfer is not synonymous with avoidable ICU transfer. Earlier, appropriate ICU transfer may improve safety and increase transfer counts. A lower count may also reflect delayed escalation, bed scarcity or case mix. Raw transfer counts cannot establish benefit or causal attribution.

2 IBM Lab 1 Business Analysis Matrix

The Business Analysis Matrix compares opportunities by impact and feasibility. This case adapts the hospital exercise to four selected initiatives and adds AI deterioration prediction. Scores below preserve the user assessment; they are not validated estimates or comparative evidence.

Opportunity	Impact out of 5	Feasibility out of 5	Strategic interpretation
EHR modernization	5	4	Prioritize foundational data quality, interoperability and clinical workflow.
AI deterioration prediction	3	3	Conditional evaluation after readiness checks; value and delivery remain uncertain.
Telemedicine services	4	4	Promising parallel option, subject to a separate service and access case.
Automated scheduling	4	4	Consider targeted implementation to address documented administrative bottlenecks.

Critical review of the original rationale

The preference for EHR modernization and scheduling is strategically coherent: EHR capability supports reliable data access, while scheduling may address administrative workload. Scheduling is not a universal prerequisite for AI or telemedicine. Market availability alone does not establish local feasibility, validated outcomes or low implementation risk.

A feasibility score of 4 for EHR modernization should be tested against migration effort, interoperability, procurement, training and workflow disruption. An AI impact score of 3 reflects uncertainty in this case rather than a finding that clinical AI has intrinsically modest value. Telemedicine needs its own demand, reimbursement and service design assessment.

Quadrant interpretation

	High impact scores 4 to 5	Lower impact scores 1 to 3
High feasibility scores 4 to 5	EHR modernization; telemedicine; scheduling	None of the selected initiatives
Lower feasibility scores 1 to 3	None of the selected initiatives	AI deterioration prediction at 3 and 3

The high versus lower split is illustrative. A score of 3 is a midpoint, so assigning AI to the lower quadrant is a display convention, not a rejection rule. Keep the numeric scores visible; a different cutoff changes the quadrant without changing the underlying assessment.

Portfolio implication

Establish the minimum EHR capabilities needed for safe data use first; full EHR replacement is not assumed necessary. Assess scheduling and telemedicine against the same local resource constraints. Reserve AI funding for a readiness gated pilot rather than allowing technology enthusiasm to displace higher priority operational needs.

3 IBM Lab 2 Digital Transformation Scenario Evaluation

This evaluation applies the healthcare scenario method to the hospital investment question: assess challenges, objectives, success criteria, enabling technologies and business requirements. The broader educational scenario concerns an integrated healthcare data ecosystem; this report narrows it to deterioration prediction.

Challenge	Critical revision	Proposed response
Technical	Missing, delayed or incorrectly mapped vital signs and laboratory data can invalidate predictions.	Validate patient identity, units, timestamps, provenance, required fields and feed failure behavior.
Organizational	Alert delivery without ownership creates duplication, gaps and alarm fatigue.	Define primary responder, backup, acknowledgement, escalation and shift handover.
Clinical	Both missed deterioration and unnecessary intervention matter; true positives also consume resources.	Review false negatives and false positives, treatment consequences and escalation appropriateness.
Workforce	Reviewing alerts may increase work even if later care becomes more efficient.	Measure response time and staff minutes; train teams and revise workload allocation.
Governance	Responsibility, privacy and model changes require explicit control.	Assign clinical ownership, restrict access, log events and freeze the pilot model.

From user observations to business implications

The initial analysis correctly highlighted data access, communication, workload and alarm fatigue. A key correction is that clinicians cannot identify true and false negatives from alerts alone: false negatives require reviewing deterioration events that occurred without a preceding qualifying alert. A false positive is defined against a prespecified endpoint and prediction horizon, not simply an alert that staff disliked or a patient who did not enter ICU.

Change management is necessary but insufficient. Response capacity, reliable interfaces and explicit escalation authority must exist. Training cannot compensate for missing data or an alert that has no available responder.

Role of business analysis

The business analyst translates stakeholder needs into testable requirements, traces them to business objectives, resolves workflow conflicts and maintains the decision log. Clinical leaders define safe response and endpoint review; IT validates integration; finance validates realizable costs. The analyst should not independently set clinical thresholds or claim model safety.

Transformation assessment conclusion

The proposed value chain is integrated observations → validated prediction → actionable notification → timely clinical assessment → appropriate care. Every link is a dependency. The pilot should test the whole pathway and identify where failure occurs, rather than treating model performance as evidence of clinical effectiveness.

4 Objectives and measures of success

A business objective states the desired organizational outcome. A success criterion specifies how that outcome will be judged. An enabling technology provides capability. A requirement specifies verifiable behavior or a constraint. A dashboard measures or presents value; it does not create savings by itself.

Business objective	Success criterion and timing	Enabling capabilities
Clinical effectiveness Improve timely recognition and response without harm.	During the 12 week pilot, assess alert to first doctor contact against the illustrative KPI 1 target and safety gates. Track time to appropriate action separately.	Integrated observations, timestamped alerts, clinical context and escalation workflow.
Operational efficiency Deliver a reliable response within staff capacity.	During live pilot weeks, keep nonactionable alert burden within the illustrative KPI 2 target; review weekly. Track acknowledgement and staff minutes as diagnostics.	Shared alert queue, ownership, acknowledgement status, escalation and event logging.
Financial sustainability Establish whether a sustainable investment case is credible.	At pilot close, update the three year model under KPI 3. A scale decision needs credible value and an approved budget, not a positive 12 week ROI claim.	Reconciled clinical and cost extracts, controlled analysis workbook and finance review.

SMART discipline

Specify the eligible population, event definition, baseline, comparator, time window, responsible owner and data source before launch. Targets in this report are illustrative decision proposals, not established clinical standards. Baseline measurement and local validation are required to determine whether they are clinically meaningful and feasible.

The earlier criterion “reduction in avoidable ICU transfers” needs independent clinical adjudication, a denominator and a comparator. It is an exploratory outcome in a short pilot, not a stand alone operational success criterion. The financial objective requires a three year horizon; 12 weeks is a measurement period for inputs and feasibility, not a substitute investment horizon.

Data and technical acceptance

Clinical informatics must document the model input specification and validate mapping against source records. Feed latency, missingness tolerances, performance targets, alert repetition rules and notification service levels must be agreed for the intended use before testing. A stale or incomplete input must trigger the approved safe behavior, rather than an apparently reassuring normal score.

Benefits outside the narrow cost model

Potential benefits include earlier appropriate care, patient experience, staff confidence and capacity release. Measure them without assuming they occur. Do not monetize them without a defensible incremental estimate; do not add capacity value and avoided variable cost if they represent the same economic benefit.

5 Financial analysis and sensitivity

All figures are hypothetical, undiscounted and stated in euros. The model holds annual activity, unit cost, effect and recurring cost constant for three years and assumes full benefit from year one. No inflation, tax, financing, residual value or implementation ramp is included.

Input	Assumption	Interpretation
Hospital activity	300 beds; 12,000 admissions annually	Context; 240 unplanned transfers imply 2.0% per admission.
Baseline transfers N	240 per year	Unplanned transfers, not 240 proven avoidable events.
Avoidable variable cost C	€1,500 per avoided transfer	Must be locally validated; excludes fixed overhead allocation.
Initial investment I	€120,000	Hypothetical implementation expenditure.
Annual recurring cost O	€90,000	Hypothetical annual total; scope requires validation.
Effect r and horizon T	20%; 3 years	Assumes 48 transfers actually avoided each year.

Scenario	Avoided per year	Annual savings	Annual net	3 year net	3 year ROI
10% conservative	24	€36,000	−€54,000	−€282,000	−72.3%
20% base	48	€72,000	−€18,000	−€174,000	−44.6%
35% optimistic	84	€126,000	€36,000	−€12,000	−3.1%

Base three year benefits are €216,000; total costs are €390,000. No payback occurs under a constant base case. At 35%, simple payback would be 3.33 years, beyond the three year horizon. Three year break even requires 36.1% reduction; annual operating break even alone requires 25.0%.

One way sensitivity around the 20 percent base case

Changed input	Annual savings	3 year net	3 year ROI
C = €1,000; O = €90,000	€48,000	−€246,000	−63.1%
C = €2,000; O = €90,000	€96,000	−€102,000	−26.2%
C = €1,500; O = €60,000	€72,000	−€84,000	−28.0%
C = €1,500; O = €120,000	€72,000	−€264,000	−55.0%

Sensitivity values are illustrative analyst scenarios. Recurring cost is varied as a whole because no software only cost split exists. At 20% effect, break even needs €2,708 per avoided transfer or recurring cost no higher than €32,000 annually, holding other inputs constant.

6 Decision and controlled pilot design

Investment decision

Reject immediate full deployment on the current narrow financial model. A controlled pilot is conditionally preferred to immediate rejection only if leadership values the learning enough to fund a capped evaluation and the hospital can meet all prerequisites. If readiness, response capacity or affordable evaluation cannot be secured, defer the pilot.

Scope and comparison

Run a 12 week pilot on two internal medicine wards with comparable wards continuing usual care. Select comparison wards prospectively using patient acuity, admissions, occupancy, staffing, case mix and historical outcome trends. Document any shared staff and changes in clinical pathways that could contaminate comparisons. Do not withhold routine care or established escalation.

Complete approvals, data mapping, retrospective local validation, baseline collection and staff training before the 12 week clock begins. An illustrative schedule is weeks 1–2 silent operation and readiness verification, weeks 3–10 supervised live use, and weeks 11–12 continued use and closeout data checks. Analyze silent and live periods separately; outcome follow up may extend beyond week 12.

Evaluation method

Prespecify endpoints, prediction horizon, eligibility, exclusions, repeat alert handling and data sources. Use timestamped logs reconciled to clinical records. For outcomes, compare rates per eligible admission and patient days, report counts and uncertainty, and examine adjusted change relative to comparison wards and the preperiod. Assess baseline comparability and trends before interpreting a difference in differences estimate.

Independent clinicians should review potential deterioration, missed alerts and whether transfers were appropriate or plausibly avoidable, using an agreed protocol and masking where feasible. Adjustment cannot remove all bias in a small nonrandomized design. Report confounding, missing data, contamination, uncertainty and protocol deviations.

Learning value and sample size

At the hypothetical hospital wide rate of 240 transfers annually, only about 55 events would occur across the whole hospital in 12 weeks. Two wards would contribute fewer events; their volume is unspecified. The pilot therefore primarily tests feasibility, response and safety processes. It cannot be assumed to establish a 20% reduction, mortality benefit or positive economic return. A statistician should assess the design and event requirements before outcome claims.

Budget and accountability

Approve a separate capped pilot budget covering licensing, interfaces, training, staff response, evaluation, security and contingency. No pilot price is assumed here. The clinical sponsor owns safety; ward leads own response; IT owns availability; the analyst owns traceability; finance owns cost validation. A steering group reviews safety weekly and makes the closeout decision.

7 Three pilot KPIs and safety gates

Every numeric threshold below is explicitly illustrative. None is an established clinical standard or a validated model acceptance limit. Local clinical, nursing, informatics and finance owners must approve or replace them before launch. Passing targets does not override a safety gate.

KPI 1 Clinical response

Definition: median minutes from qualifying live alert delivery to first documented doctor contact with the patient, across all eligible alert episodes. This is not acknowledgement or necessarily treatment. Reconcile system and clinical timestamps; report the 90th percentile, missing timestamps and episodes with no contact, which must not silently disappear from the denominator.

Illustrative decision bands: GO for further evaluation at ≤10 minutes; AMBER at >10 to ≤30 minutes; NO GO at >30 minutes. A favorable median alone cannot pass the gate: delayed or absent response to high risk alerts requires clinical review. Compare with a prespecified baseline response measure; alert based timing is not directly equivalent to recognition based timing under usual care.

KPI 2 Operational alert burden

Definition: independently reviewed nonactionable alerts per 100 occupied patient days on pilot wards during live use. “Nonactionable” means no additional action was clinically warranted at the time under the agreed review protocol; it is distinct from a model false positive. Deduplicate episodes by a prespecified rule and capture staff minutes spent reviewing and responding.

Illustrative decision bands: GO at ≤5 nonactionable alerts per 100 patient days; AMBER at >5 to ≤10; NO GO at >10. Review weekly and by shift. These proposed values replace overlapping hourly rules and normalize for exposure; they require a capacity based local check before use. Total workload, including true positives, remains a safety consideration.

KPI 3 Financial sustainability

Definition: updated three year undiscounted ROI using validated incremental costs and an explicitly extrapolated range of sustainable annual benefit. Illustrative bands: GO to consider a larger evaluation if the conservative credible case is ≥0%; AMBER if only an optimistic case reaches ≥0%; NO GO for financially justified scale up if all credible cases are <0%. If effect cannot be estimated credibly, classify as inconclusive and do not approve financial scale up.

This is a forward looking model, not measured 12 week ROI. Clinical value could support a separately funded strategic decision despite negative ROI, but that exception requires explicit leadership justification. Current 10%, 20% and 35% scenarios all fail the financial scale criterion.

Safety gates and decision hierarchy

Pause live alerts immediately for suspected system related serious harm, missed or delayed essential escalation, unsafe patient identification, critical feed failure or unauthorized access. Revert to usual care and investigate. Before restart, the clinical sponsor and relevant technical owners must document resolution. Review missed deterioration, subgroup performance, response coverage and burden weekly against locally approved limits. No automatic model retraining is allowed.

Any unresolved safety gate means STOP. All KPI targets met with no safety issue supports further evaluation, subject to evidence quality and funding. Amber or inconclusive results support redesign or a longer study. Repeated unacceptable workflow performance means stop or redesign; a short pilot is never an automatic scale authorization.

8 Stakeholder needs and business requirements

Needs describe the problem or desired outcome. Requirements specify what must be delivered and how acceptance will be demonstrated. The following consolidate the initial stakeholder analysis and add groups necessary for safe operation. Acceptance tests are proposals for local specification.

Stakeholder and need	Requirement	Acceptance evidence
Physicians Understand and act on risk with usable documentation.	Show patient, alert time, relevant observations, data age, model limits and response pathway; support documentation.	Clinicians complete representative alert and documentation scenarios; verify record linkage and context.
Nursing staff Know who owns the alert and whether action occurred.	Display owner, acknowledgement and escalation status; provide coverage and handover.	Simulate day, night, duplicate alert and no response cases; no unowned alert remains beyond the approved escalation limit.
IT and clinical informatics Maintain reliable, supportable integration.	Validate feeds, identity, units, latency, failure signaling and recovery; version interfaces.	Source to display reconciliation, failure drills and recovery tests meet approved specifications.
Hospital management and finance Understand safety, resource use and affordability.	Provide reconciled outcome and cost extracts plus a controlled business case.	Finance reproduces calculations and reconciles inputs; no dedicated dashboard needed in pilot.
Patients and representatives Receive safe, understandable and equitable care.	Provide an approved communication approach and review eligibility, access and subgroup risks.	Patient materials reviewed; concerns and subgroup findings logged and escalated.
Clinical safety and rapid response teams Retain clear accountability.	Define escalation, override, incident review and fallback.	Scenario tests and downtime drill passed; named decision authority recorded.
Privacy security and governance owners Protect data and control use.	Restrict roles, audit access and changes, define retention and approve intended use.	Role tests, audit retrieval and applicable local governance reviews completed.

Conflicts to resolve before launch

Sensitivity versus alert burden: clinical and nursing owners approve the operating threshold after local validation. Documentation completeness versus workload: agree a minimum record with optional richer integration. Financial savings versus safe escalation: clinical need takes priority over transfer reduction targets. Management transparency versus privacy: use minimum necessary access and appropriately aggregated reporting.

The initial requirement “easy integrability” becomes tested interfaces and failure behavior. “Clearly displayed” becomes a usable alert with ownership and escalation. A financial dashboard is optional; reliable financial measurement is not. Record unresolved requirements with an accountable owner and due date before live use.

9 MoSCoW requirements for the pilot

Must requirements are entry conditions for live operation. Should requirements improve the pilot and may be deferred only with an accepted safe alternative. Won't means excluded from this pilot scope, not permanently rejected. No additional Could items are required for this baseline.

ID and requirement	Priority	Minimum acceptance and rationale
R1 Clinical data integration	Must	Validate required inputs, patient identity, units, timestamps, missingness and latency. Test stale feed behavior and downtime fallback. Prediction depends on trustworthy data.
R2 Explainable alerts	Should with Must minimum	Must show patient, alert time, relevant observed data and freshness, limitations and required next step. Richer explanation is Should and must be validated; it cannot imply causal certainty.
R3 Alert acknowledgement	Must	Named owner, timestamped acknowledgement, visible status and backup escalation. Test no response and handover cases. Acknowledgement does not equal assessment or treatment.
R4 Financial analytics dashboard	Won't	Use controlled exports and a reproducible finance reviewed model. A dashboard is excluded; financial measurement remains required.
R5 EHR documentation integration	Should with Must minimum	Must safely record alert, assessment, action and disposition in the patient record through an approved workflow. Automated integration is Should; test any manual alternative for traceability and burden.
R6 Mobile push notifications	Won't conditionally	Exclude only if an existing channel reliably reaches the responsible team on every shift. Test delivery, backup and escalation. If it fails, implement a reliable channel before live use; mobile is one option.
R7 Audit trail and access control	Must	Role based access and attributable logs for access, alerts, acknowledgements, actions and configuration. Verify unauthorized access rejection, log retrieval and approved retention.
R8 Automated model retraining	Won't	Freeze and version the validated model and configuration for evaluation. Monitor performance and drift without automated updates; any change needs validation, approval and a protocol amendment.

Critical changes from the original user prioritization

R2 moves from Could to Should with a clinically necessary Must minimum. R5 moves from Could to Should while safe documentation becomes Must. R7 moves from Could to Must because accountable, controlled operation cannot be deferred. R8 moves from Should to Won't to preserve evaluation stability and change control. R1, R3 and the scope exclusions remain, with explicit safety conditions for R6.

Traceability: R1 and R2 support clinical response; R3 and R6 support reliable workflow; R5 supports continuity and evaluation; R7 supports governance; R4 preserves proportionate financial measurement; R8 protects model stability. Entry testing must cover the complete pathway, not isolated functions only.

Appendix A Financial formulas and interpretation

Reproducible model

N = annual baseline unplanned transfers; r = fraction actually avoided; C = realizable variable cost avoided per transfer; I = initial cost; O = recurring annual cost; T = years. All transfer reductions in this model are assumptions about genuinely avoided events, not directly observed causal effects.

Measure	Formula	Base result
Avoided transfers per year	N × r	240 × 20% = 48
Annual gross savings S	N × r × C	48 × €1,500 = €72,000
Annual operating balance	S − O	€72,000 − €90,000 = −€18,000
Total benefits	T × S	3 × €72,000 = €216,000
Total costs	I + T × O	€120,000 + 3 × €90,000 = €390,000
Net benefit	T × S − I − T × O	−€174,000
ROI	Net benefit ÷ total costs	−€174,000 ÷ €390,000 = −44.6%
Three year break even reduction	(I + T × O) ÷ (T × N × C)	€390,000 ÷ €1,080,000 = 36.1%
Operating break even reduction	O ÷ (N × C)	€90,000 ÷ €360,000 = 25.0%

Pilot extrapolation protocol

Do not multiply a raw 12 week ward transfer reduction by 52/12 and call the result hospital savings. First estimate comparable rates, eligible exposure and the range of plausible effect with uncertainty. Then state assumptions about hospital population, sustained adoption, implementation ramp, capacity and realizable cost avoidance. Use the same full cost scope for benefits and costs.

The pilot affects two wards, while the €120,000 and €90,000 assumptions represent the investment case. Do not assign a pro rata share of these costs to the pilot without a quotation and resource plan. Report actual pilot expenditure separately from any future hospital wide model.

Cost interpretation

Avoided variable costs can include resources genuinely no longer consumed, net of additional ward care, tests and response effort. Allocated average ICU cost is not automatically cash saving. Released beds or staff time may create capacity value without reducing spending. Incremental downstream effects, implementation overruns and evaluation costs should be assessed before approving a revised total cost model.

Appendix B Evidence register and next decision

Provenance and status

Item	Status and use
Case Studies conversation	User exercises on the issue tree, financial model, pilot KPIs, challenges, objectives, stakeholder needs and MoSCoW provide the case foundation.
IBM Lab 1 Business Analysis Matrix	Hospital opportunity exercise supplied as pasted text. Adapted to four selected initiatives; AI is a case extension. Ratings are user assessments.
IBM Lab 2 Digital Transformation Scenario Evaluation	Healthcare scenario and evaluation tasks supplied in the conversation. Applied to this hospital; no IBM endorsed investment conclusion is implied.
Financial inputs	Hypothetical assumptions preserved from the case brief. Scenario outputs and break even values are arithmetic, not empirical findings.
Revised requirements and pilot plan	Consulting synthesis and proposals. No completed implementation, validation or stakeholder approval is implied.
KPI thresholds and sensitivity inputs	Explicitly illustrative proposals, not clinical standards, validated economic forecasts or observed data.

Limitations

No local dataset, observed clinical effect, vendor quotation, ward census, validated avoidability fraction or complete cost breakdown is established. Consequently, the financial model is a simplified decision aid. The 240 unplanned transfers must not be relabeled as avoidable. A nonrandomized short pilot is vulnerable to selection bias, seasonality, case mix, staffing changes, bed availability and limited events.

No legal or regulatory compliance determination is made. The responsible hospital functions must establish applicable obligations for the intended use, patient population, data flows and product before operation. Clinical targets require local approval and must align with existing escalation standards.

Required decisions before commitment

Clinical sponsor: approve intended use, endpoint, response protocol, safety limits and review method. IT and informatics: demonstrate data readiness, reliable delivery, recovery and model version control. Nursing and medical leads: confirm staffing, training and handover. Finance: validate costs, affordability and the capped pilot budget. Governance owners: complete applicable reviews and approve data use.

Required output at pilot close

Produce a transparent report of population and exposure, missing data, model performance, all three KPIs, harms and missed events, staff burden, comparison results with uncertainty, actual pilot cost and a revised scenario model. Recommend stop, redesign, extended evaluation or a separately justified deployment decision. Do not infer causality from raw transfer counts or present extrapolated ROI as realized return.